

Rational expressions

A rational expression is a quotient of polynomials with a nonzero denominator.

$$\frac{x + 2}{x - 5}$$

Which input is excluded?

Finding restrictions

$$\frac{3x}{x^2 - 16}$$

For which real values of x is this expression undefined?

Simplifying a rational expression

$$\frac{x^2 - 25}{x^2 + 2x - 15}$$

What simplified expression is equivalent on the original domain?

Preserving the original domain

$$\frac{x^2 - 9}{x - 3}$$

After simplifying, which input must still be excluded?

Canceling factors

$$\frac{x(x+4)}{x}$$

What can we cancel, and what restriction remains?

A sum in the numerator

$$\frac{x + 4}{x}$$

Can we cancel the x terms in this expression?

Opposite factors

$$\frac{3 - x}{x - 3}$$

What constant value does this expression have wherever it is defined?

Multiplying rational expressions

$$\frac{x^2 - 4}{x^2 - 9} \cdot \frac{x + 3}{x + 2}$$

What simplified expression results, with all original restrictions recorded?

Dividing rational expressions

$$\frac{x^2 - 1}{x + 3} \div \frac{x - 1}{x + 3}$$

What simplified expression results, and which inputs are excluded?

A zero divisor

Assume x is nonzero.

$$\frac{1}{x} \div \frac{x-2}{x}$$

Why must $x = 2$ also be excluded?

Adding with a common denominator

$$\frac{3x}{x-2} + \frac{5}{x-2}$$

What is the sum, and where is it defined?

Subtracting with a common denominator

$$\frac{5x + 1}{x + 4} - \frac{2x - 3}{x + 4}$$

What is the difference, and where is it defined?

Finding a common denominator

$$\frac{2}{x} \quad \text{and} \quad \frac{3}{x+1}$$

What least common denominator could we use?

Adding with different denominators

$$\frac{2}{x} + \frac{3}{x+1}$$

What single rational expression represents this sum?

Subtracting with different denominators

$$\frac{4}{x-2} - \frac{1}{x+2}$$

What single rational expression represents this difference?

Nested denominator factors

$$\frac{1}{x-1} + \frac{2}{x^2-1}$$

What simplified rational expression represents this sum?

A repeated denominator factor

$$\frac{1}{x} + \frac{2}{x^2}$$

What single rational expression represents this sum?

Complex rational expressions

A complex rational expression contains a fraction in its numerator or denominator.

$$\frac{\frac{3}{x}}{\frac{6}{x^2}}$$

What simplified expression is equivalent for nonzero x ?

Simplifying a complex fraction

$$\frac{1 + \frac{2}{x}}{1 - \frac{4}{x^2}}$$

What simplified expression results, with all original restrictions recorded?

Combining before simplifying

$$\frac{x}{x-3} - \frac{3}{x-3}$$

What simplified value results, and which input remains excluded?

An average-cost expression

A print shop spends \$120 in setup costs and \$4 per booklet. It produces n booklets.

What rational expression gives the average cost per booklet for positive n ?